

EBN122 SEMESTER TOETS/TEST 1 (25 AUG 2016)

ANTWOORDBLAD / ANSWER SHEET

Van / Surname:		Studentenommer: Student number:							
Voorletters / Initials:		Taal / Language	Afrikaans	English (Eng 3-1)	English (Eng 3-6)				

Q1		Q2		Q3		Tot:	/45 41
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VRAAG 1 – QUESTION 1 (8)

1.1	$R = 350.4 \checkmark \checkmark$
1.2	$\cancel{R} = J = -1.25 J \checkmark$
1.3	$P_1 = -63W \checkmark \checkmark$ $P_2 = 10W \checkmark$

VRAAG 2 – QUESTION 2 (16)

2.1	$P_{(1A)} = -7.5W \checkmark$	$P_{(7\Omega)} = 4.86W \checkmark$
2.2	$V_1 = 6V \checkmark$	$I_1 = 0.9A \checkmark$
2.3	$V_x = 1V \checkmark$	$V_y = -2.5V \checkmark$
2.4	$R_{ab} = 6.4\Omega \checkmark$	
2.5	$R_{cd} = 8.21\Omega \checkmark$	

VRAAG 3 – QUESTION 3 (22)

3.1	$a = -1 \checkmark$ -3	$b = 16 \checkmark$ 22	$c = -3 \checkmark$ -1	$d = 22 \checkmark$ 16
3.2	$e = 7 \checkmark$	$f = -4 \checkmark$	$g = 15 \checkmark$	$h = -9 \checkmark$
3.3	$m = 0.67 \checkmark$	$n = 0 \checkmark$	$p = 1 \checkmark$	$q = -1 \checkmark$
3.4	$V_0 = 7.31V \checkmark \checkmark$			
3.5	$\alpha = 5 \checkmark$			
3.6	$\begin{bmatrix} 1.2 & -0.5 \checkmark & -0.2 \\ -0.5 & 1 & -0.5 \\ -0.2 & -0.5 & 0.7 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \checkmark \\ 5 \\ 10 \end{bmatrix}$			
3.7	$a) V_S = 24.67V \checkmark \checkmark$	$b) V_{CE} = 14.15V \checkmark \checkmark$		